One Policy, Infinite NPCs: Scalable Persona-Conditioned NPC Control via Shared Reinforcement Learning Policies

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***Abstract*—On a 300-persona life-simulation benchmark, PCSP achieves zero-shot persona identification up to 17× above chance, Spearman $\rho \approx 0.73$ semantic-behavioral alignment, and 22× faster inference than an LLM-as-policy baseline. Life simulation games require hundreds to thousands of non-player characters (NPCs) that behave consistently with distinct personalities while remaining controllable through designer-authored natural language. Hand-authored behavior trees, per-character RL policies, unsupervised skill discovery, and per-step LLM controllers each fail on one or more deployment constraints: persona consistency, natural-language controllability, zero-shot generalization, and real-time inference. We introduce PCSP (Persona-Conditioned Shared Policy), a single reinforcement learning policy conditioned on frozen LLM embeddings of free-form persona descriptions. PCSP combines once-per-NPC persona encoding, low-rank persona projection, neural persona conditioning, and a PPO + InfoNCE consistency + KL diversity training objective. Across three Mini-Inzoi experimental settings, including a richer 20-action v3 ontology, ablations show that the InfoNCE trajectory-consistency objective is load-bearing: removing it collapses zero-shot persona identification to chance even when task reward is preserved or improved. These results provide empirical evidence that shared RL policies can support scalable, real-time, persona-conditioned NPC control, and that trajectory traceability is central to the method.**

***Index Terms*—game AI, NPC personalization, reinforcement learning, persona conditioning, large language models, life simulation**

I. Introduction

Modern life simulation games—*The Sims*, *Animal Crossing*, *inZOI*, and emerging open-world titles—place NPCs at the center of the player experience. For these games to feel alive, each NPC must behave consistently with a distinct personality: the gregarious chef and the reclusive artist should pursue the same biological needs through recognizably different activity patterns, social approaches, and daily rhythms. At scale—hundreds to thousands of NPCs per game world—this creates a fundamental challenge that current game AI architectures were not designed to address.

A. The NPC Personalization Scaling Gap

a) Behavior trees: remain the industrial standard [1]. A skilled designer hand-crafts decision logic for each character archetype. This produces believable, predictable NPCs, but authoring cost scales linearly with character count, and trees are brittle outside authored scenarios. Ten thousand distinct NPC personalities require ten thousand hand-crafted trees.

b) Per-NPC RL: appears to offer automation: train a separate policy for each persona. In principle this allows unlimited characters with arbitrary behaviors. In practice, memory and training cost also scale linearly, and each new persona requires a full retraining cycle—a prohibitive expense for live-service games that regularly introduce new characters.

c) LLM-as-policy: methods [2]–[4] achieve rich, natural-language-grounded persona expression by querying a language model at every decision step. This works well in offline or turn-based contexts, but even our local Qwen3-1.7B LLM-as-policy baseline requires 43.7 ms per decision step (Table III), which exceeds the 16–33 ms per-frame budget of real-time game simulations. Generative Agents [2] sidestep this by operating on minute-resolution plans rather than per-frame actions, which suits narrative simulation but cannot drive moment-to-moment NPC movement and activity selection.

d) Unsupervised skill discovery: (DIAYN [5], CIC [6]) learns a shared policy conditioned on latent skill codes, achieving behavioral diversity without per-character training. But the codes carry no semantic content. A game designer cannot specify “this NPC should be agreeable and fitness-oriented” by selecting a latent index. Natural-language controllability is not part of the design.

B. Persona-Conditioned Shared Policies

We address this scaling problem with a different decomposition: *compute a rich persona representation once per NPC, then condition a single lightweight policy on that representation at every step*.

The key enabler is the modern LLM embedding space. A frozen language model maps free-form persona descriptions to dense vectors that capture semantic relationships among personalities. A shared policy conditioned on these embeddings can generalize to *any* persona a designer writes, without retraining, because the continuous embedding space provides coverage of the entire personality manifold. At inference time, the LLM is called once per NPC lifetime; the policy network—at most a few hundred kilobytes—runs at full game speed.

This approach, which we call *persona-conditioned shared policies*, remains underexplored in game AI. This paper presents a concrete implementation, PCSP, and evaluates when persona-conditioned behavior remains recoverable from generated trajectories. The central empirical question is not whether

TABLE I
PARADIGM COMPARISON. ✓ = SATISFIED, × = NOT SATISFIED, △ = PARTIALLY SATISFIED.

Paradigm	Persona consist.	NL control.	Zero-shot gen.	Real-time (<5 ms)
Behavior trees	△	×	×	✓
Goal-cond. RL	×	×	×	✓
Lang-cond. RL	△	✓	△	✓
Skill discovery	△	×	×	✓
Per-NPC RL	✓	×	×	✓
LLM-as-policy	✓	✓	✓	×
PCSP (ours)	✓	✓	✓	✓

a policy can receive persona text as input, but whether its trajectories preserve enough persona signal to support zero-shot identification, semantic alignment, and real-time deployment.

C. Paper Contributions

1) **PCSP**: A shared RL controller combining frozen Qwen3-0.6B-Embedding, low-rank persona projection, neural persona conditioning, and a PPO + InfoNCE consistency + KL diversity co-training objective (§III).
2) **Empirical results**: On a 300-persona life-simulation benchmark, PCSP achieves zero-shot persona identification up to 17× above chance, Spearman $\rho \approx 0.73$ semantic-behavioral alignment, and 22× faster inference than LLM-as-policy (§IV).
3) **Ablation finding**: Across v1, v2, and v3 environments, removing the InfoNCE trajectory-consistency loss collapses zero-shot persona identification to chance, while the best conditioning architecture depends on the out-of-distribution split.
4) **Evaluation protocol**: We report persona identification, behavioral diversity, semantic-behavioral alignment, inference latency, and rich trajectory observability as core criteria for persona-conditioned NPC control (§IV-B).

II. WHY CURRENT PARADIGMS FALL SHORT

Table I evaluates six paradigms against the four axes that matter for practical life-simulation NPC deployment.

a) Goal-conditioned RL.: UVFA [7] and HER [8] condition policies on goal *states*—what to achieve. Persona describes *how to behave*: two NPCs with identical needs but different personalities should fulfill them through different activity sequences. Goal conditioning addresses the wrong axis of variation and provides no natural-language interface.

b) Language-conditioned RL.: BabyAI [9] conditions on natural-language *instructions* that change each episode. Persona is a stable trait persisting over the NPC's lifetime; treating it as an episode-level instruction discards the long-horizon consistency constraint. Decision Transformer [10] and similar sequence-modeling approaches condition behavior on past returns or goals, but not on static identity descriptors that must generalize zero-shot.

c) Unsupervised skill discovery.: DIAYN [5] and CIC [6] achieve behavioral diversity via learned latent codes, but the codes carry no semantic content. Generalizing to a new designer-specified persona requires solving an inverse problem: find the latent code for "an introverted accountant who prioritizes learning." This is not supported by design.

d) LLM-as-policy.: Generative Agents [2], Voyager [3], and ReAct [4] produce rich, persona-consistent behavior but remain expensive when used as per-step controllers. In our benchmark, the Qwen3-1.7B LLM-as-policy baseline requires 43.7 ms per decision step (Table III), already above a 16–33 ms frame budget. Techniques such as action caching, plan reuse, and smaller distilled models can reduce latency, but the fundamental tension—rich language reasoning versus real-time step budgets—remains. Generalist agents such as Gato [11] demonstrate that a single model can handle diverse tasks, but inference cost is similar.

e) Summary.: No existing paradigm satisfies all four axes. The design space between "fast but no NL control" and "NL control but too slow" has not been systematically explored. Persona-conditioned shared policies are a concrete proposal for how to close this gap.

III. PCSP METHOD

We present PCSP as one specific point in the design space of persona-conditioned shared policies. The components we have chosen are motivated by ablation evidence but are not canonical; they define the experimental system evaluated below.

A. Environment: Mini-Inzoi

Mini-Inzoi is a PettingZoo AEC life-simulation environment on a 6×6 grid with 4 agents, 8 bio-social needs (hunger, sleep, social, leisure, hygiene, fitness, work, learning), and 12 discrete actions (8 activity + 4 movement). Observation $s \in \mathbb{R}^{20}$: position (2), time-of-day (1), needs (8), and summaries of the other three agents (9). Reward combines needs satisfaction, persona-aligned activity bonuses (+0.5 for preferred actions), and social bonuses scaled by Big Five compatibility [12] ($0.2 + 0.3 \times \text{cos-sim}(\text{BF}_i, \text{BF}_j)$).

Persona dataset. 300 persona texts are generated from 15 Big Five archetypes × 20 occupations (240 train / 60 zero-shot test). Each text is a 2–3 sentence natural-language description that an NPC designer might write for a new character.

B. Persona Encoding

Given persona text p, we compute a frozen LLM embedding $\hat{e}_p \in \mathbb{R}^{1024}$ using Qwen3-0.6B-Embedding [13] (last-token pooling, L2-normalized). This is computed once per NPC lifetime (14.6 ms/persona), imposing negligible runtime cost.

A learned low-rank projection [14] ($r = 16$, $\text{lr} = 10^{-4}$) maps $\hat{e}_p \to e_p \in \mathbb{R}^{64}$:

$$e_p = \text{norm}(\alpha\, BA\hat{e}_p)\,,$$
$$A \in \mathbb{R}^{16\times1024}, \quad B \in \mathbb{R}^{64\times16}, \quad \alpha = 64^{-1/2}.$$

The projection is necessary: raw Qwen3 embeddings over-represent occupational similarity relative to personality similarity (salesperson↔executive: 0.61 vs. trainer↔blogger: 0.33).

After low-rank projection training, Spearman ρ between projected persona distance and behavioral KL increases from 0.384 to 0.728 (§IV).

C. Persona-Conditioned Shared Policy

The shared policy $\pi_\theta(a \mid s, e_p)$ is a 3-hidden-layer MLP (256-256-128) with an output head sized to the environment action space. Our reference implementation uses FiLM [15] to inject the persona signal at every hidden layer:

$$h'_\ell = \gamma_\ell(e_p) \odot h_\ell + \beta_\ell(e_p),$$

with small linear networks $\gamma_\ell, \beta_\ell : \mathbb{R}^{64} \to \mathbb{R}^{d_\ell}$, where d_ℓ is the hidden width of layer ℓ. An analogous FiLM-conditioned value network $V_\psi(s, e_p)$ serves as the critic. Total trainable parameters: 207K (excluding the frozen LLM). We treat the choice of conditioning mechanism as an implementation detail rather than a contribution: the v3 zero-shot ablations in §IV show that input-concatenation conditioning is competitive with FiLM and in fact generalizes more strongly to unseen occupations. The load-bearing component is the consistency objective introduced next, which is what makes persona signal recoverable from trajectories regardless of how the policy is conditioned.

D. Co-Training Objective

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda_1 \mathcal{L}_{\text{consist}} + \lambda_2 \mathcal{L}_{\text{diverse}}, \tag{1}$$

with $\lambda_1 = 0.5$, $\lambda_2 = 0.1$ (PPO: $\gamma = 0.99$, $\epsilon = 0.2$, GAE-$\lambda = 0.95$, batch $= 2048$ [16]).

Consistency loss. A 2-layer GRU trajectory encoder $g_\eta(\tau) \in \mathbb{R}^{64}$ maps episode trajectories to L2-normalized embeddings. InfoNCE [17] ($T = 0.07$) aligns trajectory embeddings with their originating persona embeddings:

$$\mathcal{L}_{\text{consist}} = -\log \frac{\exp\big(\text{sim}(g_\eta(\tau), e_p)/T\big)}{\sum_{p' \in \mathcal{B}} \exp\big(\text{sim}(g_\eta(\tau), e_{p'})/T\big)}. \tag{2}$$

This prevents the policy from producing trajectories that cannot be traced back to their conditioning persona.

Diversity loss. We maximize the expected KL divergence between policy distributions induced by different personas at shared states:

$$\mathcal{L}_{\text{diverse}} = -\mathbb{E}_{s,\, p \neq p'}\big[D_{\text{KL}}\big(\pi_\theta(\cdot \mid s, e_p) \,\|\, \pi_\theta(\cdot \mid s, e_{p'})\big)\big], \tag{3}$$

computed as a batched forward pass over $n = 8$ sampled personas at 32 states. This discourages behavioral collapse toward a persona-agnostic average.

IV. Experiments and Results

We evaluate PCSP across three settings. Mini-Inzoi v3 is the primary experiment because it exposes a richer action ontology for persona-conditioned behavior. Mini-Inzoi v1 validates the base architecture at small scale, and Mini-Inzoi v2 tests whether the semantic-behavioral alignment pattern persists under a larger grid, more agents, and more personas.

A. Experimental Setup

v3 (richer action ontology). Mini-Inzoi v3 keeps the 6×6/4-agent base scale but expands the action space to **20 flat-discrete actions** (16 activity + 4 movement) for finer behavioral granularity (e.g., `focused_work` vs. `planning_work`, `rest_alone` vs. `rest_with_others`, `eat_quick` vs. `eat_slow`). Observations expand to 33 dimensions, adding location-affordance one-hot (8), social-context features (3), and a routine-regularity signal (2). Reward shaping adds a per-action persona-style term $r_{\text{style}} = 0.3 \cdot \cos(\mathbf{p}_{\text{BF}}, \mathbf{s}_a)$ that aligns each action with a hand-authored Big-Five style profile. Same 240/60 unseen-occupation split as v1; 300 PPO iterations.

v1 (base scale). Mini-Inzoi 6×6, 4 agents, 300 personas (240 train / 60 zero-shot test). The held-out 60 personas comprise *four entire occupations* (HR manager, professor, data scientist, chef) crossed with all 15 Big Five archetypes, so this evaluates **unseen-occupation compositional generalization**: the target occupation never appears in training, only the personality structure does. All models train for 300 PPO iterations (≈1.9M steps) on an NVIDIA RTX 6000 Ada (49 GB VRAM). Baselines: **B1** No-Persona PPO; **B3** SBERT-conditioned policy (`all-MiniLM-L6-v2`, 384-dim, frozen); **B4** DIAYN (random 64-dim); **B5** LLM-as-policy (Qwen3-1.7B).

v2 (expanded scale). Mini-Inzoi **12×12, 16 agents, 500 personas** (400 train / 100 zero-shot test), 200 PPO iterations. Same hyperparameters and evaluation protocol as v1. The observation dimension expands to 56 (position 2, time 1, needs 8, 15 other agents × 3). Random zero-shot chance is 1.0% (1/100 test personas).

B. Evaluation Protocol

We evaluate persona-conditioned NPC control with metrics that separate task performance from persona fidelity.

a) Persona identification accuracy.: Given a trajectory, an automated k-NN classifier identifies which held-out persona generated it. We report zero-shot accuracy on held-out personas, with random chance determined by the number of test personas.

b) Behavioral diversity.: Mean pairwise KL divergence between action distributions conditioned on different persona embeddings measures whether policies collapse to persona-agnostic behavior despite high task reward.

c) Semantic-behavioral alignment.: Spearman ρ between projected persona distances and pairwise behavioral KL divergences measures whether semantic distances in persona space are reflected in behavior space. The projected embeddings are L2-normalized, so Euclidean distance is monotonic with cosine distance. High ρ with low KL variance can indicate alignment without meaningful diversity, so both quantities are reported where available.

d) Inference latency.: We report GPU ms/step at batch size 1. Real-time game integration typically requires per-step control below a 16–33 ms frame budget.

e) Rich trajectory observability.: Identification and believability scores depend on what a trace exposes. Coarse action sequences can compress out temporal, spatial, social,

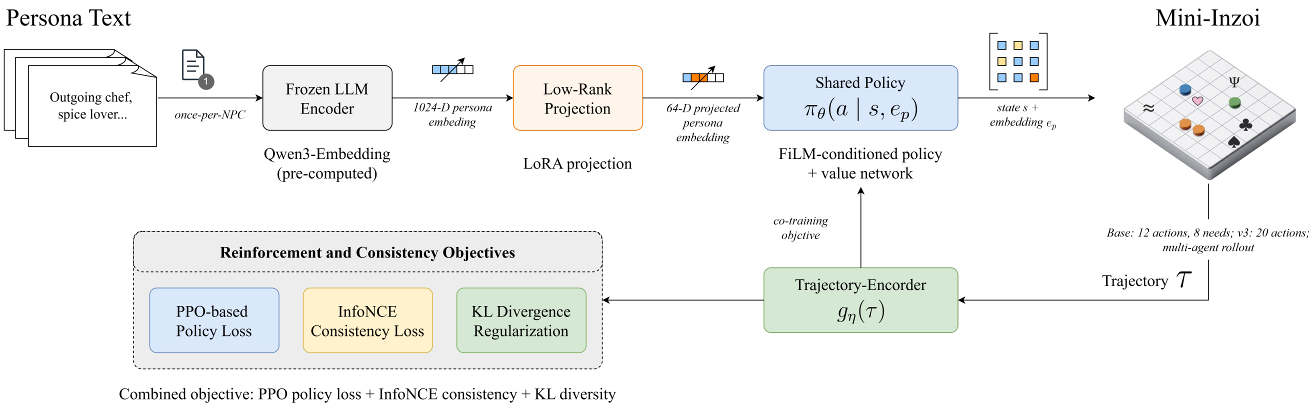

Fig. 1. **PCSP pipeline and training objectives.** Persona text is encoded once per NPC with a frozen Qwen3 embedding model, adapted through a low-rank projection, and consumed by a shared persona-conditioned policy during Mini-Inzoi rollout. The trajectory encoder provides the InfoNCE consistency signal, while the policy is optimized with PPO and KL diversity regularization. The rightmost Mini-Inzoi label depicts the base 12-action/8-need instantiation; the v3 experiments expand the action ontology to 20 actions.

TABLE II
V3 RESULTS (6×6, 4 AGENTS, 300 PERSONAS, 20-ACTION ONTOLOGY; 240 TRAIN / 60 ZERO-SHOT HELD-OUT OCCUPATIONS). ZS ACC: ZERO-SHOT k-NN ACCURACY ON 60 UNSEEN-OCCUPATION PERSONAS (RANDOM ≈1.7%); BRACKETS ARE WILSON 95% CIS. COH.: TRAJECTORY COHERENCE RATIO (INTRA/INTER PERSONA COSINE SIMILARITY).

Model	Reward ↑	ZS Acc ↑	Coh. ↑
PCSP (full)	104.1	0.170 [.13,.22]	2.06
PCSP (no_consist)	118.4	0.017* [.01,.04]	1.07
PCSP (no_diverse)	122.1	0.160 [.12,.21]	2.05
PCSP (concat)	107.0	**0.283** [.24,.34]	**7.60**
B1 No-Persona	100.3	—	—
B3 SBERT	106.9	—	—

* Near-random (failure mode).

TABLE III
V1 RESULTS (6×6, 4 AGENTS, 300 PERSONAS). ZS ACC: ZERO-SHOT k-NN ACCURACY ON 60 UNSEEN PERSONAS (RANDOM ≈1.7%). ρ: SPEARMAN (PROJECTED PERSONA DIST. VS. KL).

Model	Reward ↑	ZS Acc ↑	ρ ↑	Lat. ↓
PCSP (full)	83.4	**0.193**	**0.728**	1.97 ms
PCSP (no_consist)	84.3	0.017*	0.638	2.03 ms
PCSP (no_diverse)	76.8	0.147	—†	1.79 ms
B1 No-Persona	73.2	—	—	1.83 ms
B3 SBERT	**86.2**	—	—	1.88 ms
B5 LLM-policy	—	—	—	43.7 ms

* Near-random (failure mode). †KL ≈0.39; ρ not meaningful.

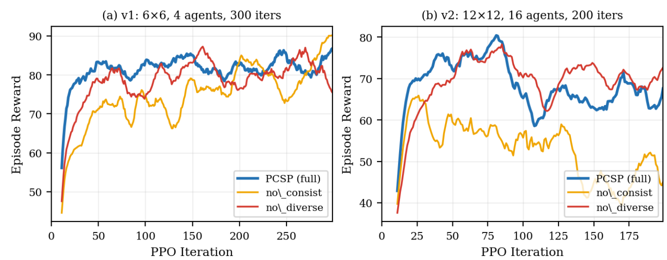

Fig. 2. **Training reward curves at two scales.** (a) v1 (6×6, 4 agents, 300 iters). (b) v2 (12×12, 16 agents, 200 iters). Removing the consistency loss (no_consist) preserves or even slightly increases reward but collapses zero-shot persona traceability across all scales (Tables III–II). Removing the diversity loss (no_diverse) causes KL collapse at v1 and v2 (0.39 and 0.48) and a v1 reward drop, but its effect on v3 zero-shot accuracy is within the Wilson CI of full (§IV).

TABLE IV
V2 RESULTS (12×12, 16 AGENTS, 500 PERSONAS). ZS ACC ON 100 UNSEEN PERSONAS (RANDOM =1.0%). LOWER ABSOLUTE REWARD REFLECTS THE HARDER, LARGER ENVIRONMENT.

Model	Reward ↑	ZS Acc ↑	ρ ↑	KL ↑
PCSP (full)	64.2	0.023	**0.725**	**5.40**
PCSP (no_consist)	50.1	0.000*	0.253	16.47‡
PCSP (no_diverse)	71.6	0.013	—†	0.48
PCSP (concat)	**74.8**	**0.027**	—†	1.34

* Exactly 0 (total failure). †KL too low; ρ not meaningful. ‡High KL without consistency reflects unstructured divergence, not persona alignment.

and stylistic dimensions that personality traits modulate. We therefore render rollouts with time-of-day, location, nearby agents, action style, and short-form event semantics for human-facing evaluation artifacts, and treat coarse-versus- rich trace comparison as part of the evaluation protocol.

C. Results

Tables II, III, and IV show key metrics for each setting. Figure 2 shows training reward curves at v1/v2.

D. Key Observations

Across all three experimental settings, the ablation evidence converges on a single load-bearing component—the InfoNCE consistency loss—while the choice of conditioning architecture proves to be secondary.

The consistency loss is the load-bearing component. Removing the InfoNCE consistency term collapses zero-shot persona identification to chance in every setting we have tested:

TABLE V
CONDITIONING ARCHITECTURE × OOD SPLIT (V3). ZERO-SHOT k-NN ACCURACY ON THREE COMPOSITIONAL HELD-OUT SPLITS, EVALUATED AGAINST THE SAME TRAINED PCSP CHECKPOINT FAMILY (RANDOM $\approx$1.7%). WILSON 95% CIS IN BRACKETS; **BOLD** MARKS THE HIGHER POINT ESTIMATE PER ROW, AND † MARKS ROWS WHERE WILSON INTERVALS OVERLAP.

OOD split	FiLM	Concat
Unseen occupations	0.170 [.13,.22]	**0.283** [.24,.34]
Unseen archetypes	**0.203** [.16,.25]	0.103 [.07,.14]
Unseen occ. × archetype †	**0.203** [.16,.25]	0.170 [.13,.22]

†Wilson intervals overlap; gap not significant. Architecture preference flips with the OOD axis: concat generalizes more strongly when the occupation is novel, FiLM when the archetype is novel.

v1 drops from 19.3% to 1.7% (random), v2 drops from 2.3% to exactly 0% against a 1.0% baseline, and v3 drops from 17.0% to 1.7% (Wilson CI $[.01, .04]$). The reward axis hides this failure entirely. At v1 the no_consist ablation even slightly *increases* reward (84.3 vs. 83.4); at v3 the gap is sharper still (118.4 vs. 104.1, +14 reward) while accuracy collapses by 15.3 percentage points. Without the consistency objective, the policy chases reward through persona-agnostic strategies whose trajectories are no longer traceable back to the conditioning persona, regardless of how the persona signal is injected into the network.

The conditioning architecture is secondary, and conditional on the type of OOD shift. We initially conjectured that FiLM's layer-wise gating would dominate input concatenation, but the three v3 compositional splits in Table V show no universal winner. Concat dominates when the held-out axis is occupation; FiLM dominates when the held-out axis is Big-Five archetype; on the joint occ. × archetype split the two are statistically indistinguishable. Thus the conditioning mechanism is a generalization-gap knob whose best setting depends on *which* dimension of the persona is novel at test time. The consistency loss, in contrast, is the only component that universally fails across split families when removed.

Diversity loss prevents behavioral collapse at v1/v2 but its effect weakens on the v3 zero-shot accuracy axis. Removing the KL diversity term reduces mean policy KL by an order of magnitude at v1 (5.87 to 0.39, 15×) and v2 (5.40 to 0.48, 11×), matching the original behavioral-collapse story. At v3 zero-shot, however, diversity-loss removal lands within the Wilson CI of full (16.0% vs. 17.0%): the policy still achieves persona-recoverable behavior on the accuracy axis without the explicit KL term. We report diversity as a useful regularizer for KL-style behavioral spread rather than as a co-equal contribution to zero-shot identification.

Spearman ρ is preserved across scales. PCSP (full) achieves $\rho = 0.728$ in v1 and $\rho = 0.725$ in v2—a difference of 0.003 despite a 4× increase in grid area, 4× more agents, and 67% more personas. This suggests that semantic-behavioral alignment is a stable property of the design, not an artefact of the minimal environment.

E. Coarse-Trace Human Pilot

To test whether the policy's persona signal is visible to human readers, we ran a 30-participant Google Forms pilot using the coarse Korean 2AFC survey. Each item showed a PCSP-v3 trajectory as a sequence of coarse action labels and asked participants to choose which of two persona descriptions generated it. Because the form preserved only item-level A/B selection ratios, we report aggregate forced-choice accuracy rather than participant-level variance, confidence, or response-time analyses.

Participants selected the correct persona in 612 of 900 aggregate judgments (68.0%; pooled Wilson 95% CI $[64.9, 71.0]$), above the 50% chance baseline. Item-level results were mixed: 15 of 30 items were strongly readable ($\geq 80\%$ correct), 2 were moderately readable (70–79%), 8 were ambiguous (40–69%), and 5 were misleading ($< 40\%$). Thus coarse traces do expose some persona-conditioned behavior, but they leave a substantial minority of items ambiguous or inverted. We interpret this as evidence for the observability bottleneck motivating richer trajectory rendering, not as a completed rich-versus-coarse human comparison.

F. Qualitative Case Study: Designer-Authored Personas

To test whether PCSP can be controlled by persona descriptions outside the synthetic Big-Five/occupation templates, we constructed 50 designer-authored personas drawn from five sources at varying distance from the training distribution: *The Sims 3* trait combinations [18] (*e.g.* Workaholic + Ambitious + Perfectionist; $n = 10$), *Animal Crossing* villager personality categories [19] (*e.g.* Lazy, Jock, Normal; $n = 12$), *Stardew Valley* NPC archetypes (*e.g.* the reclusive coder, the off-grid hermit; $n = 10$), *Persona*-series confidant descriptions (*e.g.* the perfectionist student leader, the hikikomori hacker; $n = 6$), and original designer briefs authored from scratch from Big-Five composites (*e.g.* burned-out doctor, sleep-deprived startup founder; $n = 12$). Each source concept was rewritten as a 2–3 sentence natural-language NPC description and assigned a plausible occupation. All 50 designer-authored persona texts are written in *English*, while the 240 training personas are Korean; the case study therefore additionally serves as a fully cross-lingual robustness probe of the frozen multilingual Qwen3 embedding and the learned low-rank projection. No retraining was performed: each text was encoded with the same Qwen3 embedding pipeline, passed through the learned low-rank projection, and rolled out for five v3 episodes.

Outcome and failure taxonomy. Each rollout is classified by overlap between its top-3 actions and the authored preferred-action signature. Outcomes are *success* (overlap ≥ 2), *partial* ($= 1$), or *failure* ($= 0$). Failures are further tagged by mechanism: **F1** ontology gap (authored behavior is not expressible in the 20-action space, *e.g.* "protect," "nurture"), **F2** style-reward conflict (top-3 mean style-cosine with the persona Big-Five vector is negative), **F3** embedding occupational bias (rollout follows the nearest train-occupation neighbor rather than the authored personality), **F4** trait collision (a declared multi-trait

TABLE VI
DESIGNER-AUTHORED PERSONA CASE STUDY (50 ENGLISH PERSONAS, 5 SOURCES, KOREAN-TRAINED POLICY). OUTCOME BREAKDOWN FROM FIVE V3 ROLLOUTS PER PERSONA; FAILURE-MODE COLUMN LISTS THE DIAGNOSES PRESENT IN THAT SOURCE'S FAILURE CASES. MEAN COSINE SIMILARITY TO THE NEAREST (KOREAN) TRAINING PERSONA IS 0.590 (RANGE 0.478–0.686).

Source	n	Succ.	Part.	Fail.	Failure modes
The Sims 3	10	4	5	1	F1
Animal Crossing	12	4	5	3	F1, F2, F5
Stardew Valley	10	5	4	1	F1
Persona series	6	4	2	0	—
Original brief	12	5	5	2	F2, F4
Total	**50**	**22**	**21**	**7**	**86% non-failure**

composite collapses to a single trait), and **F5** residual (zero overlap with none of the above triggers).

Three patterns are worth highlighting. First, 43 of 50 personas (86%) produce at least one top-3 action matching the authored intent, and 22 (44%) match at least two—substantially above the chance baseline of matching 2 of 3 from a 20-action set ($\approx 9\%$). This holds even though every designer-authored text is in English while the trained policy has only ever seen Korean persona descriptions, providing direct evidence that the multilingual Qwen3 backbone plus the learned low-rank projection yields a language-agnostic persona representation. Figure 3 makes the role of the projection explicit: in the raw Qwen3 space the English designer personas form a clearly separated cluster (12.9$\times$ the train–train nearest-neighbor distance), but after the learned LoRA projection that the policy actually consumes they are fully intermixed with the Korean training personas (1.47$\times$). The mean cosine similarity between each English persona and its nearest Korean training neighbor drops only modestly (0.608 with the Korean-text variant of the same designer set vs. 0.590 here), and per-source success rates are stable or higher across the language shift.

Second, source-stratified results expose a meaningful gradient: Stardew Valley NPC archetypes generalize best (50% strong / 40% partial / 10% failure), The Sims 3 trait combinations and original Big-Five briefs sit at a similar level (40% / 50% / 10% and 42% / 42% / 17%), and the Persona-series confidants never fail outright (67% / 33% / 0%). Animal Crossing personality categories remain the hardest source (25% failure), concentrated in F1, F2, and F5 tags. This is consistent with the AC categories being defined almost entirely by stylistic and affective markers (*e.g.* Cranky, Snooty, Sisterly) that the v3 ontology cannot distinguish at the action level.

Third, the dominant failure mechanism is ontology-limited rather than embedding-limited. Of seven failures, five (F1$\times$3, F4, F5) involve composite or stylistic concepts the 20-action ontology cannot distinguish: the *normal villager* (hygiene/care behavior, F1), *family-oriented caretaker* (protective family role, F1), *animal clinic worker* (gentle care semantics, F1), *sleep-deprived startup founder* (multi-trait collision, F4), and the *jock villager* (fitness-coded social action collapses into the generic `socialize_initiate` bucket, F5). Only two failures are squarely policy-side: the *cranky villager* and the *restless rideshare driver* (both F2, the per-action style reward outranks the persona-conditioned intent). Notably, the F3 embedding-occupational-bias failure that appeared in the Korean-text variant disappears here—with English designer text and Korean training neighbors, the embedding can no longer latch onto a shared occupation token. This mirrors the coarse-action observability problem discussed in §VI and is the cleanest argument in this work that the next environment iteration should expand the social/stylistic axis of the action space rather than the embedding or training pipeline.

G. Cross-Scale Replication

The absolute task reward and zero-shot accuracy are lower in v2, as expected for a substantially harder environment with more test personas. The same empirical structure is observed across environment scales: the InfoNCE consistency objective is required for zero-shot persona traceability, and the semantic-behavioral alignment property ($\rho \approx 0.73$) is preserved. Figure 4 shows the zero-shot distribution for v1; the v2 pattern is qualitatively similar with narrower per-persona bars. The aggregate evaluation gives nearly identical semantic-behavioral alignment across environments ($\rho = 0.728$ vs. 0.725). Figure 5 visualizes an independent empirical sample of persona pairs from the same checkpoints, again showing strong monotone alignment at both scales.

V. DISCUSSION AND FUTURE WORK

Dynamic personas and memory. Current PCSP encodes a static persona text once at NPC creation. Life-simulation characters should also change: a repeated social conflict may increase neuroticism, a long friendship may shift agreeableness, and a remembered event may alter future action preferences. A natural extension is to separate stable persona traits from dynamic memory state, using a lightweight event-conditioned update model or retrieval-augmented memory module that can influence the shared policy without requiring per-step LLM inference [2].

Scaling to richer environments. Mini-Inzoi isolates persona-conditioned control in a deliberately small grid world. The next environment target is a richer multi-agent benchmark such as Melting Pot [20], where social dilemmas, partial observability, and larger interaction spaces can test whether persona-conditioned policies produce stable social patterns rather than only different marginal action frequencies. Engine integration also remains a practical step: the policy must serialize game state, tolerate asynchronous events, and preserve the sub-frame inference profile observed in the Python benchmark.

Human evaluation methodology. Automatic metrics are necessary for ablations but insufficient for perceived NPC believability. Our 30-participant coarse-trace pilot shows above-chance persona identification, but also many ambiguous or misleading items. Stronger human evaluation will require an instrument that preserves per-participant responses, so that

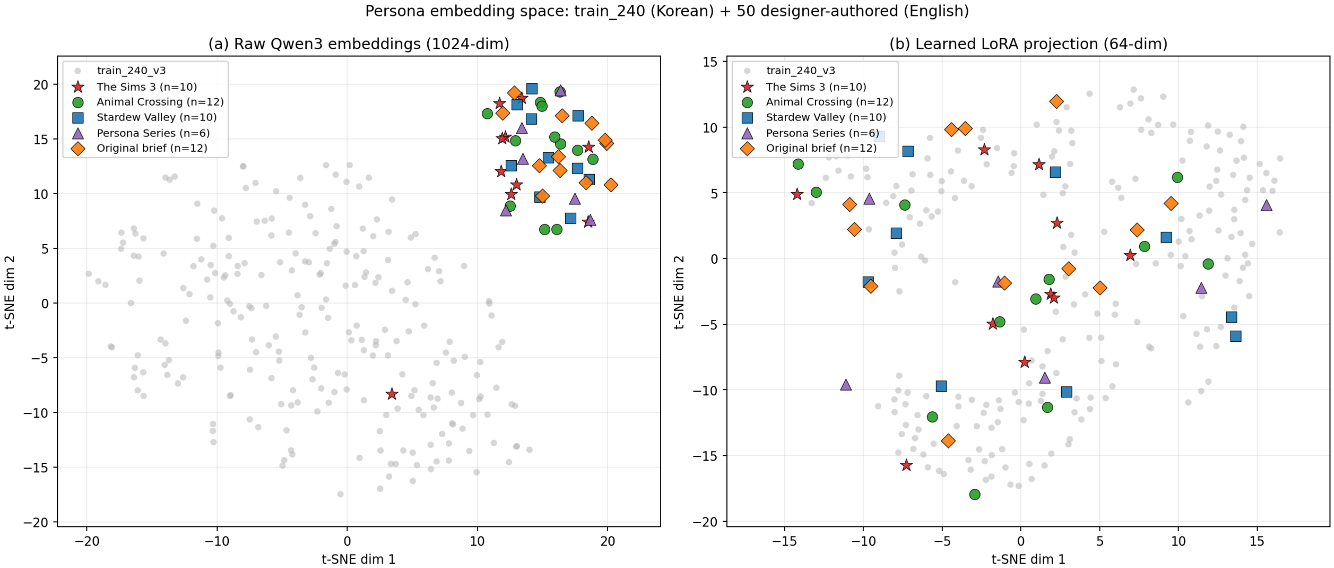

Fig. 3. **Designer-authored personas in embedding space: raw vs. learned projection.** t-SNE of 240 Korean training persona embeddings (grey) plus 50 English designer-authored personas, color-coded by source. **(a)** In the raw 1024-dim Qwen3 embedding space, the English designer personas form a language-shifted cluster well outside the Korean training distribution: their mean nearest-neighbor distance to any training persona is $12.9\times$ the mean train–train nearest-neighbor distance. **(b)** After the learned 64-dim LoRA projection used by the policy, the same designer personas are fully intermixed with training personas across all five sources (ratio drops to $1.47\times$). The projection is what makes the case-study setting effectively in-distribution for the trained policy, and is the load-bearing piece of the cross-lingual robustness reported in §IV-F.

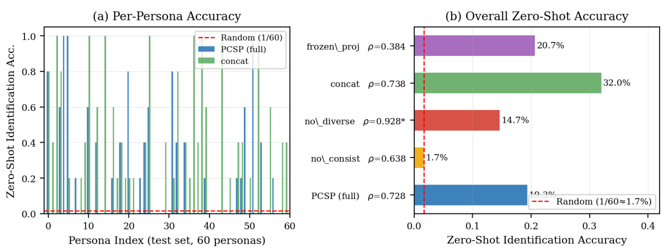

Fig. 4. **Zero-shot generalization on 60 unseen personas (v1).** PCSP (full) achieves 19.3% trajectory-to-persona identification; the red dashed line marks random chance (1.7%). Removing the consistency loss collapses accuracy to 1.7%. The v2 experiment (100 unseen personas) replicates this qualitative pattern.

variance, confidence, and inter-rater agreement can be analyzed alongside aggregate accuracy.

VI. Limitations and Scope

PCSP has several limitations that bound the interpretation of the results.

Minimal environments. Mini-Inzoi v1–v3 are far smaller than commercial life-simulation worlds. The v2 experiment tests a larger grid and more agents, but continuous physics, longer horizons, asynchronous events, and richer social affordances remain untested.

Action observability. The original 12-action space exposes only a thin slice of persona-relevant behavior. In the 30-participant coarse-trace pilot, aggregate 2AFC accuracy was above chance, but 13 of 30 items were ambiguous or misleading. Coarse labels such as `read`, `rest`, and `eat` are difficult to

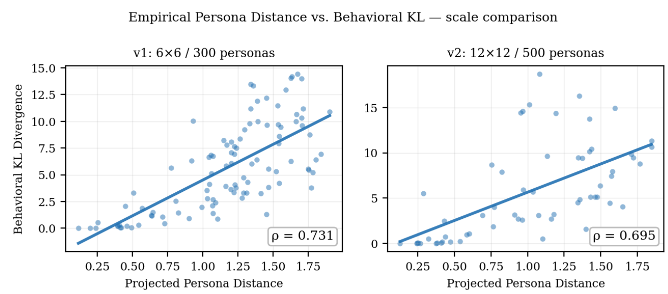

Fig. 5. **Empirical projected persona distance vs. behavioral KL divergence at two scales (PCSP full).** Each point is a recomputed persona pair from the trained policy checkpoints rather than a synthetic reconstruction from summary statistics. Left: v1 sampled-pair $\rho = 0.755$ (100 pairs, 200 states). Right: v2 sampled-pair $\rho = 0.717$ (60 pairs, 100 states). The monotone alignment between persona space and behavior space is preserved when moving to a $4\times$ larger grid with $4\times$ more agents and 67% more personas.

interpret without temporal, spatial, and social context. Mini-Inzoi v3 and the rich rollout renderer address this partly, but finer social and stylistic action semantics remain needed.

Synthetic personas. Training personas are generated from Big Five archetypes and occupation templates. The designer-authored case study in §IV-F tests 50 additional handwritten personas across five sources, but does not establish robustness to the nuance, inconsistency, or cultural specificity of production-authored characters.

Limited aggregate human evaluation. The completed Google Forms pilot collected only item-level A/B selection ratios for coarse traces. It supports aggregate 2AFC accuracy analysis, but not participant-level variance, confidence, response time, order effects, or inter-rater reliability. A more instrumented human study remains future work.

No engine integration. The policy runs in a Python simulation environment. Integration challenges with commercial game engines remain unknown.

VII. Conclusion

Life simulation games create a scaling challenge that the game AI field has not yet systematically addressed: how to give hundreds to thousands of NPCs distinct, consistent, and controllable personalities without proportional authoring or inference cost. We show that *persona-conditioned shared policies*—a single RL policy conditioned on frozen LLM persona embeddings—can satisfy the four axes that matter for practical deployment: persona consistency, natural-language controllability, zero-shot generalization, and real-time inference.

Across three Mini-Inzoi settings, PCSP achieves up to $17\times$ above-chance zero-shot persona identification, $\rho \approx 0.73$ semantic-behavioral alignment, and $22\times$ faster inference than an LLM-as-policy baseline. The main empirical finding is that the InfoNCE trajectory-consistency objective is load-bearing: removing it collapses zero-shot persona traceability to chance even when reward improves. The next steps are to couple static personas with dynamic memory, move the evaluation to richer multi-agent environments such as Melting Pot, and complete human studies using rich trajectory traces.

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